Helios Grid: Modular Commercial Battery Storage

Synthetic demonstration document - fictional company and product

Abstract

This whitepaper describes Helios Grid, a fictional modular battery storage platform designed by the invented company Northwind Energy Labs. It is synthetic demonstration content for NEXUS AI and does not describe a real product.

1. Problem

Commercial buildings waste between 12 and 18 percent of on-site solar generation because production peaks at midday while demand peaks in the early evening. Helios Grid stores midday surplus and discharges it between 5 pm and 9 pm.

2. Architecture

Each Helios Grid cabinet contains eight lithium iron phosphate modules of 6.4 kWh, for a usable capacity of 51.2 kWh per cabinet. Up to twelve cabinets can be chained, giving a maximum installation of 614.4 kWh. A cabinet weighs 540 kg and occupies 0.9 square metres of floor space.

3. Performance

In a 14-month pilot at three sites, round-trip efficiency averaged 91.3 percent. The pilot reduced peak grid demand by 37 percent and cut annual electricity costs by an average of 22 percent. The battery management system kept cell temperatures between 18 and 32 degrees Celsius.

4. Safety

The modules use lithium iron phosphate chemistry, which has a higher thermal runaway threshold than nickel manganese cobalt cells. Every cabinet includes aerosol fire suppression, per-module fuses and a sealed IP55 enclosure.

5. Pricing and warranty

List price is 19,800 US dollars per cabinet before installation. Northwind Energy Labs offers a 10-year warranty guaranteeing at least 80 percent of original capacity, or 6,000 full cycles, whichever comes first.

6. Limitations

Helios Grid is not designed for off-grid residential use, and the pilot sites were all in temperate climates. Performance in regions with sustained temperatures above 40 degrees Celsius has not been measured.